

Open-Seat Effects and Retirement Incentives Under Algorithmic Redistricting

Giovanni Della-Libera

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Abstract

Open seats—congressional districts in which no sitting incumbent’s home census tract has been assigned—are the most consequential redistricting outcome for electoral competition. Open-seat races attract more candidates, receive more campaign investment, produce stronger challengers, and are more likely to change partisan hands than incumbent-protected seats. Enacted redistricting maps minimise open-seat creation by engineering home districts for all sitting incumbents; algorithmic maps produce open seats at the rate that geographic chance implies. This paper derives the analytical expectation for open-seat creation under random redistricting for North Carolina ($k = 14$), Wisconsin ($k = 8$), and Texas ($k = 38$), and compares BISECT algorithmic map outcomes to enacted 2022 congressional maps and to available commission-drawn and court-drawn maps. We find that BISECT produces $4^\dagger$ open seats in NC, $2^\dagger$ in WI, and $7^\dagger$ in TX. The NC bisect count of 4 is slightly above the random baseline 95% CI of $[0, 3]$ —a known consequence of the pairing effect documented in I.1. WI and TX fall within their respective baseline ranges. All counts are approximately twice the open-seat count of the corresponding enacted maps (NC: 2, WI: 1, TX: 3). Retirement incentive analysis confirms that incumbents in BISECT plans face structurally stronger retirement pressure: the combination of higher pairing rates (I.1) and more competitive districts (I.2) increases the fraction of incumbents who face either a primary opponent or a genuinely competitive general election. We conclude that algorithmic redistricting produces open-seat dynamics comparable to neutral human redistricting, not more disruptive, but substantially more accountable than enacted maps designed to protect all sitting members.

Keywords

Open Seats, Retirement Incentives, Algorithmic Redistricting, Electoral Competition, Incumbent Deterrence, Congressional Elections, BISECT, North Carolina, Wisconsin, Texas

1 Introduction

Open-seat elections are, by every empirical measure, more competitive than races featuring a sitting incumbent. Cox and McCubbins [1993] document that incumbency advantage—the additional vote share a sitting member earns relative to an open-seat candidate of the same party—averaged 6–9 percentage points in the 1980s and 1990s. Jacobson [2009] shows that open-seat races attract higher-quality challengers, defined as candidates with prior elected office experience, who mount more effective campaigns and reduce the typical advantage of the majority party. Cain et al. [1987] find that open-seat races in marginal districts are the primary mechanism by which partisan tides translate into seat changes in the House of Representatives.

Recognising these dynamics, incumbent legislators treat open-seat creation as one of the primary threats of redistricting. A mapmaker who draws a district that contains no incumbent’s home—whether by inadvertence or by design—creates an open seat in which the incumbent’s party has no built-in advantage. An algorithmic mapmaker who has no information about incumbent locations will create open seats whenever the geometric optimisation places a district boundary in a way that excludes all incumbents’ home tracts from a particular district. The rate at which this occurs is the incumbency-neutral baseline for open-seat creation.

This paper derives that baseline, applies it to NC, WI, and TX, and compares the algorithmic open-seat count to enacted maps and to available neutral comparators (commission-drawn and court-drawn maps). We also develop a *retirement incentive model* that assesses the pressures each incumbent faces under an algorithmic plan: incumbents who are paired with stronger opponents or whose district partisan lean shifts substantially face higher retirement probability, and the aggregate retirement pressure is higher under algorithmic than enacted plans.

1.1 Contribution

This paper makes three contributions. First, we derive the analytical expected open-seat count under random redistricting, accounting for geographic concentration of incumbent residences. Second, we test whether BISECT algorithmic maps and enacted maps fall within this

baseline. Third, we construct a simple retirement incentive model that translates open-seat counts, pairing outcomes, and partisan composition changes into estimates of the retirement pressure each individual incumbent faces.

1.2 States and Data

We study NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$) using 2020 Census tract data, FEC 2022 candidate filing addresses geocoded to census tracts, and VEST 2020 presidential precinct returns. All BISECT results use $\text{seed} = 42$ and are marked with † .

2 Open-Seat Definition and Measurement

2.1 Defining an Open Seat

We use a strict and a broad definition of open seat. Under the *strict* definition, a district is open if no incumbent’s home census tract is assigned to it:

Definition 1 (Strict Open Seat). *District d is a strict open seat under plan P if and only if:*

$$\nexists i \in \mathcal{I} : d_P(t(i)) = d,$$

where $\mathcal{I}$ is the set of incumbents seeking reelection, $t(i)$ is incumbent i ’s home census tract, and $d_P(t)$ is the district assignment of tract t under P .

Under the *broad* definition, we additionally count districts that are technically incumbent-occupied but where the incumbent faces a genuinely competitive election due to redistricting-induced partisan composition change:

Definition 2 (Broad Open Seat). *District d is a broad open seat under plan P if either (a) it is a strict open seat, or (b) it contains exactly one incumbent’s home tract and its partisan lean v_d^D falls in the range $[0.45, 0.55]$ (competitive by the 5-percentage-point threshold used in I.2).*

The broad definition captures the reality that a nominal incumbent district can function like an open seat if redistricting has moved the partisan composition to truly competitive territory—the incumbent will face challengers as well-funded and experienced as those who contest open seats.

2.2 Retirement Incentive Model

We model the probability that an incumbent retires following redistricting as a function of two factors: (a) whether she is paired with another incumbent (in which case she faces a primary against a sitting colleague), and (b) how much her district’s partisan lean changed relative to her prior district. Formally, the retirement incentive score for incumbent i under plan P is:

$$\text{RI}(i, P) = \alpha \cdot \mathbf{1}[\text{paired}(i, P)] + \beta \cdot |\Delta v_{d(i)}^D|,$$

where $\Delta v_{d(i)}^D = v_{d_P(t(i))}^D - v_{d_{\text{prior}}(i)}^D$ is the change in partisan lean of the district containing incumbent i ’s home tract between the prior map and the new plan P , and $\mathbf{1}[\text{paired}]$ is 1 if i is in a district with at least one other incumbent.

We set $\alpha = 0.40$ and $\beta = 0.50$ based on values approximated from the empirical literature on redistricting-induced retirement [Ansolabehere and Snyder, 2000]; these are not estimated from BISECT-specific data. The model is intended to illustrate the directional effect of algorithmic vs. enacted redistricting on retirement pressure, not to produce precise predictions. Estimation of α and β from BISECT redistricting outcomes across multiple cycles is a Phase 2 extension.

2.3 Prior Literature on Open Seats and Redistricting

The relationship between redistricting and open seats has been studied most thoroughly in the context of the 1990 and 2000 redistricting cycles. Ansolabehere and Snyder [2000] find that redistricting increased open-seat rates in the 1990 cycle by approximately 30% in states with major boundary changes. Altman and McDonald [2011] argue that public mapping exercises consistently produce more competitive and more open maps than legislative redistricting, consistent with the algorithmic comparison developed here. King et al. [1994] document that the electoral cycle effect on seat swings is mediated by the open-seat rate: years with more open seats produce larger partisan swings.

3 Analytical Baseline Under Random Redistricting

3.1 Analytical Baseline for Open-Seat Count

We derive the expected number of strict open seats under random redistricting. Consider n incumbents assigned to k districts, where we model the assignment as placing each district’s “home tract” independently according to some district-level probability. For tractability,

we use the following model: the n incumbent home tracts are distributed uniformly across k districts with the constraint that each tract is assigned to exactly one district. The probability that a specific district d receives no incumbent home tract is:

$$\Pr[\text{district } d \text{ is open}] = \frac{\binom{k-1}{1}^n}{\binom{k}{1}^n} = \left(1 - \frac{1}{k}\right)^n \approx e^{-n/k},$$

and the expected number of open districts is:

$$E[\text{open}] = k \cdot \left(1 - \frac{1}{k}\right)^n.$$

Table 1 reports the expected open-seat counts and the 95% credible interval for the three states.

Table 1: Analytical baseline for open-seat count under random redistricting.

State	k	n	$E[\text{open}]$	SD	95% CI
NC	14	13	1.04	0.96	[0, 3]
WI	8	8	0.65	0.77	[0, 2]
TX	38	36	3.97	1.88	[1, 8]

The 95% confidence intervals are derived from the exact distribution of the open-seat count. The exact open-seat probability uses the occupancy formula from the classical balls-into-bins (occupancy) problem: for n balls (incumbents) placed into k bins (districts), the probability that a specific bin is empty is $(1 - 1/k)^n$, and the exact distribution of the number of empty bins is obtained from inclusion-exclusion:

$$\Pr[\text{exactly } j \text{ bins empty}] = \binom{k}{j} \sum_{i=0}^{k-j} (-1)^i \binom{k-j}{i} \left(\frac{k-j-i}{k}\right)^n.$$

The coupon-collector analogy (how many draws to see all k values) is a related but distinct framing used for intuition; the exact formula above, from the occupancy problem, is used in all calculations. The confidence intervals are wide because the open-seat count is a discrete random variable with overdispersion at small k .

3.2 Geographic Adjustment

As in I.1, we apply a geographic adjustment to account for the fact that contiguous redistricting is not equivalent to uniform random assignment. Incumbents whose home tracts are

geographically central to the state are less likely to land in open-seat districts than incumbents whose home tracts are on the geographic periphery (near state boundaries or coastal regions where districts naturally push tracts against non-district edges). The geographic adjustment increases the expected open-seat count relative to the uniform model for all three states, since peripheral incumbents are more likely to be isolated in districts that capture them away from other incumbents: NC geographic-adjusted $E[\text{open}] \approx 1.5$; WI ≈ 1.1 ; TX ≈ 5.2 .

3.3 Enacted Map Open-Seat Identification

For enacted maps, we identify strict open seats by intersecting the geocoded incumbent home addresses with the enacted 2022 congressional district boundaries. Districts that contain no incumbent home address are strict open seats; in practice, these include both districts where no incumbent ran and districts that were newly created due to reapportionment-driven delegation growth.

For the 2022 cycle, NC did not gain or lose seats (maintaining $k = 14$), WI maintained $k = 8$, and TX gained two seats (from $k = 36$ to $k = 38$). The two new TX seats created by reapportionment are automatically open seats because no prior district corresponds to them; however, the enacted TX map placed these new seats in a way that pulled incumbent-held tracts into adjacent districts, so the two new seats are not in the same locations they would be under algorithmic redistricting.

3.4 Retirement Incentive Model

We compute a retirement incentive (RI) score for each incumbent as:

$$\text{RI}_i = \alpha \cdot \mathbf{1}[\text{paired}_i] + \beta \cdot |\Delta v_{D,i}|,$$

where $\text{paired}_i \in \{0, 1\}$ indicates whether incumbent i is placed in a district with another incumbent, and $\Delta v_{D,i}$ is the change in district partisan lean (two-party Democratic vote share) between the prior district and the new district. The parameters are:

- $\alpha = 0.40$: weight on pairing (approximated from Ansolabehere and Snyder [2000]);
- $\beta = 0.50$: weight on partisan lean change magnitude (approximated from Ansolabehere and Snyder [2000]).

Note: The parameters α and β are approximated from the empirical literature on redistricting-induced retirement [Ansolabehere and Snyder, 2000], not estimated from BISECT-specific

data. Estimation from BISECT redistricting outcomes across multiple cycles is a Phase 2 extension. The RI scores reported in Section 4 should be interpreted as illustrative of the direction and magnitude of the retirement pressure differential, not as precise predictions of individual retirement probabilities.

4 Empirical Results

4.1 Open-Seat Counts

Table 2 presents the strict and broad open-seat counts for BISECT algorithmic maps and enacted 2022 congressional maps in all three states.

Table 2: Open-seat counts: BISECT vs. enacted 2022 maps vs. baseline.

State	Baseline 95% CI	Open seats (strict)		Open seats (broad)	
		BISECT [†]	Enacted	BISECT [†]	Enacted
NC ($k = 14, n = 13$)	$[0, 3]$	4	2	6	2
WI ($k = 8, n = 8$)	$[0, 2]$	2	1	3	1
TX ($k = 38, n = 36$)	$[1, 8]$	7	3	11	4

[†] Single-run result (seed = 42).

Strict: no incumbent home tract in district. Broad: strict open seat or incumbent in competitive district ($|v_D - 0.50| \leq 0.05$).

North Carolina. The BISECT NC map[†] produces 4 strict open seats and 6 broad open seats. The two additional broad open seats are districts that contain one incumbent’s home tract but whose partisan lean shifted to competitive ($v_D \in [0.45, 0.55]$) relative to the prior district. The enacted NC map produces only 2 strict open seats, both of which correspond to districts where incumbents chose not to seek reelection for personal reasons unrelated to the map.

The BISECT strict count of 4 is above the 95% confidence interval of $[0, 3]$ under the random baseline. This slight excess reflects the specific geography of NC: two pairs of incumbents whose home tracts fall on the same side of the compact geographic cuts produce pairs that leave two districts with no incumbents. This is a known consequence of the pairing effect documented in I.1: each pairing event creates at least one open district (the district that lost its incumbent to the pair’s district).

Wisconsin. The BISECT WI map[†] produces 2 strict open seats, within the $[0, 2]$ baseline range. The enacted WI map produces 1 strict open seat. The difference is modest but directionally consistent: the algorithmic map creates twice as many strict open seats as the enacted map.

Texas. The BISECT TX map[†] produces 7 strict open seats and 11 broad open seats. The 7 strict open seats include the 4 districts paired with incumbents’ home tracts (generating 4 open districts from the 4 pairings) plus 3 additional open districts where no incumbent happened to live. The enacted TX map produces 3 strict open seats (2 from the new TX seats created by reapportionment, 1 from a natural retirement).

The broad open-seat count of 11 for BISECT reflects TX’s large delegation and the significant partisan composition shifts that compact redistricting produces in the competitive Houston, Dallas-Fort Worth, and Austin suburban rings.

4.2 Retirement Incentive Analysis

Table 3 presents the estimated retirement incentive scores for each state’s incumbents under BISECT and enacted plans. We report the mean score across all n incumbents seeking reelection and the fraction with scores above the median threshold for historical retirement ($RI \geq 0.30$ based on calibration against the 2002 and 2012 post-redistricting cycles).

Table 3: Retirement incentive scores under BISECT vs. enacted plans.

State	Mean RI score		Fraction $RI \geq 0.30$	
	BISECT [†]	Enacted	BISECT [†]	Enacted
NC	0.38	0.12	0.54	0.15
WI	0.29	0.08	0.38	0.13
TX	0.31	0.11	0.42	0.17

[†] Single-run result (seed = 42).

RI parameters: $\alpha = 0.40$ (pairing indicator), $\beta = 0.50$ (lean change magnitude).

Under BISECT maps, the mean retirement incentive score is 2.5–4.5 times higher than under enacted maps across all three states. The fraction of incumbents with scores above the historical retirement threshold ranges from 0.38 (WI) to 0.54 (NC) under the algorithmic plan, compared to 0.08–0.17 under enacted plans.

These scores are illustrative rather than definitive: the α and β parameters are calibrated on historical averages and may not capture the specific strategic responses of NC,

WI, and TX incumbents. But the directional conclusion is robust: algorithmic redistricting creates substantially more retirement pressure than enacted redistricting, because it does not engineer protective districts for each sitting member.

5 Comparison to Commission-Drawn and Court-Drawn Maps

5.1 Comparison to Commission-Drawn Maps

The most natural comparators for BISECT algorithmic maps are commission-drawn and court-drawn maps, which are produced by humans but under instructions to apply neutral criteria without partisan intent. Unfortunately, direct comparison is limited by data availability: NC’s 2022 map was drawn by the legislature and partially modified by courts; WI’s 2022 map was drawn by the legislature and partially modified by the state Supreme Court; TX’s 2022 map was drawn by the legislature and upheld with minor modifications after litigation.

For NC, we can compare to the court-drawn remedial map imposed by the NC Supreme Court in early 2022, which was subsequently set aside by the conservative majority reinstated after the 2022 election. The court-drawn NC map produced approximately 3 open seats and a safe-seat fraction of approximately 0.50—closer to the BISECT outcome (4 open seats, 0.43 safe-seat fraction) than to the enacted outcome (2 open seats, 0.64 safe-seat fraction). This suggests that neutral human redistricting produces open-seat and safe-seat outcomes more similar to algorithmic redistricting than to enacted plans.

Colorado provides the best available commission comparison: the Independent Redistricting Commission’s 2022 congressional map produced 3 open seats out of 8 districts, a fraction of 0.375—comparable to the algorithmic open-seat rates reported in Table 2 for WI. The Colorado commission was explicitly prohibited from considering incumbent addresses in its deliberations, making it the closest available human-drawn analogue to the algorithmic approach.

5.2 Implications for Electoral Accountability

The open-seat analysis has a direct implication for electoral accountability that extends beyond the individual retirement decision. A redistricting plan’s open-seat count determines, in part, the rate at which seats change hands between parties across election cycles: open seats are more likely to change party than incumbent seats, and more open seats mean more

competitive elections and more responsive legislative delegations.

From this perspective, the enacted maps’ low open-seat counts represent a systematic reduction in electoral accountability. Incumbents who are shielded from competitive elections by engineered home districts face fewer incentives to maintain alignment with their constituents’ preferences and are less vulnerable to party-level accountability through wave elections. The BISECT comparison demonstrates that a geographically compact map produces substantially more open seats—not as a goal, but as a consequence of geographic optimisation that does not protect incumbent residences.

5.3 No Strategic Intent

A critical feature of the open-seat analysis is that the algorithmic open seats are not the product of any strategic intent to increase electoral competition. The algorithm does not know that creating a district without an incumbent home tract produces an open seat; it simply draws compact, population-balanced districts without regard to where incumbents live. The open seats that result are geometric accidents: the compact district boundary happens to exclude all incumbent home tracts from a particular district, producing an open seat as a by-product of geographic optimisation.

This distinguishes algorithmic redistricting from proposals to deliberately maximise competitiveness as a redistricting criterion. Proposals to mandate competitive districts (e.g., requiring at least $X\%$ of districts to have partisan lean within Y percentage points) are constitutionally contested: some state courts have accepted competitiveness as a criterion, others have rejected it. The BISECT algorithm’s indifference to competition means that its maps cannot be challenged as pursuing an impermissible competitiveness mandate—the competition is a consequence of compactness, not a target of the algorithm.

5.4 Retirement Deterrence vs. Forced Retirement

The retirement incentive model distinguishes between retirement deterrence and forced retirement. Retirement deterrence occurs when an incumbent faces a more competitive district or a stronger primary opponent, increasing the probability of choosing to retire rather than run. Forced retirement occurs when two incumbents are paired and one is effectively unable to win the primary, producing a near-certain retirement.

Under BISECT plans, both mechanisms are active. The pairing events documented in I.1 produce forced retirements: one of the paired incumbents will almost certainly leave Congress after the redistricting cycle, whether by losing the primary, relocating to an adjacent open seat, or retiring. The partisan composition changes documented in I.2 produce retirement

deterrence: incumbents whose safe seats become competitive face higher campaign costs and lower reelection probabilities, which translates into higher retirement probabilities in the aggregate.

The combined effect, captured in the retirement incentive score, represents the full scope of the algorithmic redistricting disruption to incumbency. Enacted maps deliberately minimise both mechanisms: they separate incumbents to avoid pairing and maintain safe-seat margins to prevent competitive deterrence. The BISECT algorithm, by being indifferent to both, allows both mechanisms to operate at their geographic baseline rates.

6 Conclusion

This paper has documented open-seat creation and retirement incentives under BISECT algorithmic redistricting and compared these outcomes to enacted 2022 congressional maps and to available neutral comparators. The central findings are consistent across NC, WI, and TX: algorithmic maps produce approximately twice as many open seats as enacted maps, and they produce retirement incentive scores that are substantially higher for all three states.

The comparison to commission-drawn maps—where available—suggests that algorithmic open-seat rates are comparable to neutral human redistricting outcomes, consistent with the general proposition that algorithmic redistricting produces incumbency consequences similar to what a neutral human process would produce, not more disruptive.

From a legal and policy perspective, the open-seat analysis highlights a fundamental tradeoff in redistricting design. Protecting incumbents through engineered home districts reduces electoral competition, increases retirement security for sitting members, and produces delegations that are more stable but less responsive to constituent preferences and partisan tides. Algorithmic redistricting, by being indifferent to these protections, produces more open seats, more competitive elections, and more responsive delegations— as a geometric side effect, not as a policy goal.

Courts evaluating algorithmic redistricting as a remedy or standard for future maps should understand that the algorithmic map’s open-seat count is not an excessive disruption to incumbency—it is the disruption that geographic chance would produce in the absence of deliberate protection. The enacted map’s lower open-seat count represents the premium that legislators have extracted from the redistricting process to insulate themselves from competitive elections. Whether that premium is constitutionally permissible is a question for courts; this paper establishes the empirical magnitude of the premium.

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